

Consider a partially ordered set P .

Call a nonempty partially ordered set $D \subseteq P$ *countably directed* if every countable subset of D has an upper bound in D .

Call $k \in P$ $\aleph_1$ -**compact** if for every countably directed $D \subseteq P$

$$k \leq \bigvee D \implies k \leq d \text{ for some } d \in D.$$

Lemma. Consider set S and its powerset $\mathcal{P}(S)$ partially ordered by $\subseteq$.

Consider a Scott continuous $f : \mathcal{P}(S) \rightarrow \mathcal{P}(S)$.

Consider $\text{Fix}(f) = \{x \subseteq S : f(x) = x\}$, $\text{Fix}(f)$ being partially ordered by $\subseteq$ as well.

Then for every $x \in \text{Fix}(f)$, $x = \bigvee \{k \leq x : k \text{ is } \aleph_1\text{-compact}\}$.

Remark: There is **no cardinality restriction on S** in this Lemma.

Proof.

Fact. Directed joins in $\text{Fix}(f)$ are unions.

This immediately follows from Scott continuity of f : For every directed $D \subseteq \text{Fix}(f)$, $\bigcup_{d \in D} d = \bigcup_{d \in D} f(d)$, and $f(\bigcup_{d \in D} d) = \bigcup_{d \in D} f(d)$, therefore $\bigcup_{d \in D} d$ is a fixed point of f , therefore $\bigcup_{d \in D} d \in \text{Fix}(f)$.

Now fix $A \in \text{Fix}(f)$ and a token $a \in A \subseteq S$. Scott continuity gives

$$f(A) = \bigcup_{u \subseteq_{\text{fin}} A} f(u).$$

Because $A = f(A)$, the formula above means that for each $b \in A$ we can choose a finite witness $u_b \subseteq A$ such that $b \in f(u_b)$.

Starting with $B_0 = \{a\}$, define

$$B_{n+1} = B_n \cup \bigcup_{b \in B_n} u_b, \quad B = \bigcup_{n < \omega} B_n.$$

Each B_n is finite, so B is countable. (**Note:** countable means either finite or countably infinite.) Every token in B has its chosen witness inside B ; consequently,

$$a \in B \subseteq f(B), \quad B \subseteq A.$$

We can therefore iterate upwards:

$$K = \bigcup_{n < \omega} f^n(B).$$

Scott continuity implies $f(K) = K$. Moreover, $K \subseteq A$, and **K is the least fixed point containing B .**

To see that K is $\aleph_1$ -compact, suppose $D \subseteq \text{Fix}(f)$ is countably directed and

$$K \subseteq \bigvee_{\text{Fix}(f)} D = \bigcup D.$$

The countable set B is contained in the union of countably many members of D . Countable directedness supplies one $d \in D$ containing all of B . Since d is fixed and K is the least fixed point containing B ,

$$K \subseteq d.$$

Thus every token of A belongs to an $\aleph_1$ -compact fixed point below A , proving the Lemma.

Astra gave the following remark: Notice that K **itself need not be countable**. Its compactness comes from its countable generating set B .

The reason why K might not be countable is that f is generally allowed to map a finite or countably infinite subset of S to a subset of S which has higher cardinality.